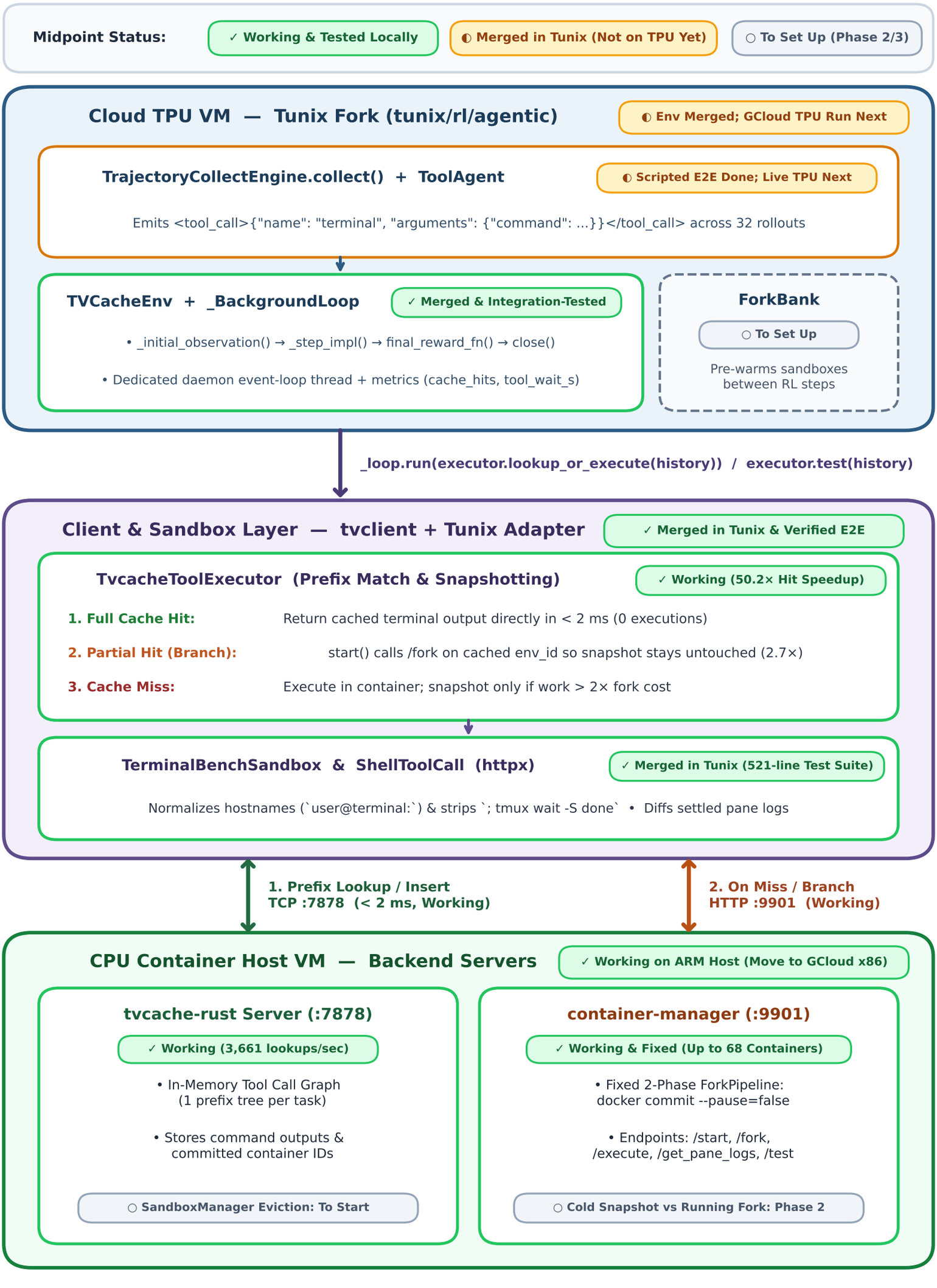

Midpoint Status:
✓ Working & Tested Locally
◐ Merged in Tunix (Not on TPU Yet)
○ To Set Up (Phase 2/3)
Cloud TPU VM — Tunix Fork (tunix/rl/agentic)
◐ Env Merged; GCloud TPU Run Next
TrajectoryCollectEngine.collect() + ToolAgent
◐ Scripted E2E Done; Live TPU Next
Emits <tool_call>{"name": "terminal", "arguments": {"command": ...}}</tool_call> across 32 rollouts
TVCacheEnv + _BackgroundLoop
✓ Merged & Integration-Tested
• _initial_observation() → _step_impl() → final_reward_fn() → close()
• Dedicated daemon event-loop thread + metrics (cache_hits, tool_wait_s)
ForkBank
○ To Set Up
Pre-warms sandboxes between RL steps
_loop.run(executor.lookup_or_execute(history)) / executor.test(history)
Client & Sandbox Layer — tvclient + Tunix Adapter
✓ Merged in Tunix & Verified E2E
TvcacheToolExecutor (Prefix Match & Snapshotting)
✓ Working (50.2× Hit Speedup)
1. Full Cache Hit: Return cached terminal output directly in < 2 ms (0 executions)
2. Partial Hit (Branch): start() calls /fork on cached env_id so snapshot stays untouched (2.7×)
3. Cache Miss: Execute in container; snapshot only if work > 2× fork cost
TerminalBenchSandbox & ShellToolCall (httpx)
✓ Merged in Tunix (521-line Test Suite)
Normalizes hostnames (`user@terminal:`) & strips `; tmux wait -S done` • Diffs settled pane logs
1. Prefix Lookup / Insert
TCP :7878 (< 2 ms, Working)
2. On Miss / Branch
HTTP :9901 (Working)
CPU Container Host VM — Backend Servers
✓ Working on ARM Host (Move to GCloud x86)
tvcache-rust Server (:7878)
✓ Working (3,661 lookups/sec)
• In-Memory Tool Call Graph (1 prefix tree per task)
• Stores command outputs & committed container IDs
○ SandboxManager Eviction: To Start
container-manager (:9901)
✓ Working & Fixed (Up to 68 Containers)
• Fixed 2-Phase ForkPipeline: docker commit --pause=false
• Endpoints: /start, /fork, /execute, /get_pane_logs, /test
○ Cold Snapshot vs Running Fork: Phase 2